

D.K.T.E Society's Textile and Engineering Institute, Ichalkaranji

Smart Surveillance Framework for Agriculture Motor and Cable Theft Detection

Synopsis of Proposed Work Submitted to

**Department of
Electronics and Telecommunication Engineering**

For the Degree of

Bachelor of Technology

In

Electronics and Telecommunication Engineering

By

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**DKTE Society's Textile and Engineering Institute, Ichalkaranji
(An Autonomous Institute)**

Utility Patent Filed: "Agriculture Motor Theft and Cable Tampering Detecting Smart System with GPS, GSM, Gyroscope and Heal" | **Filed:** 7th December 2025 | **Application No.:** 202521123445

1. Introduction

Agriculture is the backbone of rural life, and water motor pumps are one of the most essential tools for irrigation, ensuring timely water supply to crops. These motors are often installed in remote farm locations, which makes them highly vulnerable to theft of both the motor and the copper wiring. Farmers not only face direct financial losses due to stolen equipment but also suffer from disrupted irrigation schedules, which can lead to crop damage and reduced productivity.

India has over 20 million irrigation pump sets, making this a large-scale, urgent problem with no affordable smart solution currently available. While this issue is critical in agriculture, wire theft is not limited to farms alone. Similar incidents are common in railway signaling cables, streetlight wiring, and power transmission lines, causing large-scale disruptions and financial losses across the country.

The proposed FarmSecureX system addresses this by developing an intelligent, low-cost surveillance and protection system using a novel coded pulse continuity technique, GPS, GSM, gyroscope, and camera module to provide real-time monitoring, instant alerts, and location tracking.

2. Motivation

From the farmer's perspective, both theft and unexpected motor failures cause repeated financial and operational difficulties. A stolen motor or cut wire means irrigation comes to a halt, and arranging repairs or replacements often takes days, during which crops may dry up. Many farmers report recurring losses due to copper theft, rewiring expenses, and frequent motor breakdowns.

The motivation behind this project is to provide a reliable, farmer-friendly system that ensures both security and real-time response. With SMS and voice call alerts via GSM, Google Maps GPS tracking, gyroscope-based theft classification, and ESP32-CAM visual evidence capture, farmers can respond quickly to theft attempts. This dual functionality not only protects farmers from theft but also reduces downtime and improves irrigation reliability.

3. Literature Survey

Ref.	Paper Details	Key Findings
01	Kottur farmers lament frequent theft of copper wire used for agricultural motor-pump sets (The Hindu)	Farmers face repeated copper theft leading to financial losses and irrigation disruptions. Stolen wires sold in scrap market.
02	Copper Cable Theft: Revisiting the Price-Theft Hypothesis	Copper theft is price-driven and specifically motivated by market value, not broader economic trends.
03	Antitheft Alarm System for Electric Motor Pumps (Thorat & Rathod, 2017)	Low-cost anti-theft system using PIC microcontroller and MAX485. FarmSecureX

		improves on this with GPS, GSM, gyroscope, and coded pulse technique.
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Gap Analysis from Literature:

- Most research targets urban/utility copper theft, not rural agricultural theft.
- Available systems are costly or complex, unsuitable for small farmers.
- No integrated solutions combining coded pulse sensing, GPS, GSM, gyroscope, and camera evidence capture.
- Scarcity of real-world validated anti-theft systems with proven live test results for agriculture.

4. Problem Statement

Develop a smart surveillance framework equipped with sensors, GSM/GPS communication, and a coded low-voltage continuity interface to provide real-time alerts, reliable monitoring, and scalable protection for agriculture motor and cable theft detection in remote rural environments.

5. Objectives

1. Develop a coded low-voltage continuity sensing channel (200ms pulse) for cable tampering detection.
2. Implement real-time detection and classification of both cable theft and motor theft scenarios.
3. Integrate Gyroscope, GSM, and GPS modules for instant alerts and Google Maps location tracking.
4. Incorporate ESP32-CAM for visual evidence capture and email delivery during theft events.
5. Incorporate a rechargeable battery backup system for uninterrupted off-grid operation.
6. Provide SMS + voice call dual-alert user interface for remote notifications and monitoring.

6. Core Innovation — Coded Pulse Technique

Heart of the System: The Raspberry Pi Pico W generates a unique coded pulse signal every 200ms through the motor cable at low voltage. This coded pattern is continuously verified at the receiver end.

How it works:

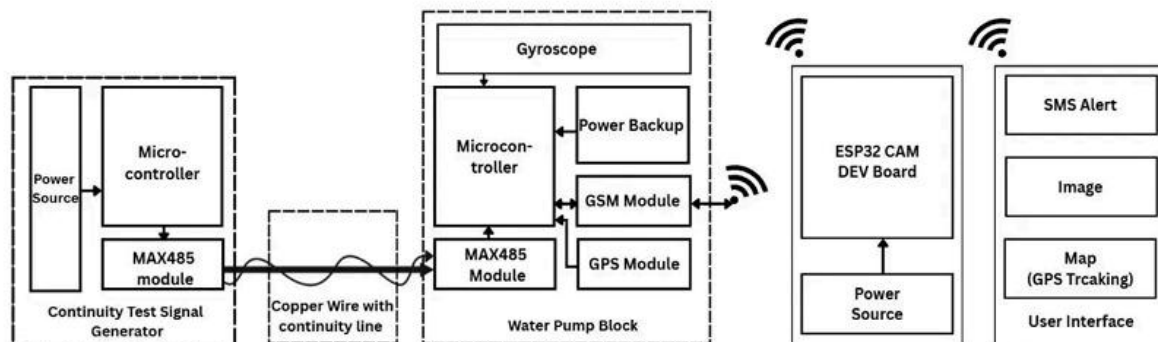
- If cable is intact: coded pulse received and verified — no alert
- If cable is cut or tampered: signal loss or mismatch detected — wire tamper alert triggered
- If wire tampered + gyroscope detects vibration: motor theft alert triggered (most critical)

This approach is UART-free, noise-immune, and far more reliable than conventional threshold-based methods. It accurately distinguishes between cable-only theft and motor+cable theft scenarios.

7. System Architecture & Block Diagram

The system consists of two hardware units:

- Continuity Test Signal Generator: Microcontroller + MAX485 + Power Source — generates coded pulses
- Water Pump Block: Microcontroller + Gyroscope + GSM + GPS + MAX485 + Power Backup + ESP32-CAM



8. Hardware Components

Component	Purpose
Raspberry Pi Pico W	Main controller — generates coded pulses every 200ms
SIM A7670E (GSM+GPS)	SMS alerts + voice calls + GPS location tracking
GY-GPS6MV2	GPS module for accurate coordinates
MPU6050 (Gyroscope)	Vibration/motion detection for motor theft classification
ESP32-CAM (OV2640)	Visual evidence capture at UXGA 1600x1200 resolution
RS485 / MAX485 Module	Robust long-distance coded signal transmission
Rechargeable Battery	Off-grid continuous 24/7 operation

9. Working Principle

7. Raspberry Pi Pico W generates coded pulses every 200ms through motor cable
8. Receiver continuously verifies the unique coded pattern
9. If cable is intact: signal verified, no alert triggered
10. If cable is cut/tampered: signal loss or mismatch detected — Wire Tamper Alert sent
11. Gyroscope simultaneously checks if motor vibration is present
12. System classifies: Cable theft only OR Motor + Cable theft

13. GSM module sends SMS alert + Google Maps GPS link to farmer
14. Voice call placed automatically after 3 second delay
15. ESP32-CAM captures photo evidence and emails via SMTP

10. Alert Classification

Scenario	Trigger Condition	Action Taken
Wire Tampering	Signal loss or mismatch detected	SMS + Call + 1 Photo
Motor Theft	Wire tampered + vibration detected (> 4.0 deg/s)	SMS + Call + 3 Photos
Normal Operation	Coded pulse verified every 200ms	No alert

11. Live Test Results

All results measured from live hardware testing on actual deployed system:

Parameter	Result
Wire tamper detection time	< 10 seconds
Motor theft detection time	< 1 second
SMS alert delivery	Successful
Voice call	Completed after 3s delay
Gyroscope threshold	4.0 deg/s
Motor theft magnitude (test)	4.5 deg/s (confirmed)
Photo resolution (ESP32-CAM)	UXGA 1600x1200
Photo file size	~856 KB (JPEG)
SMTP email delivery	100% success
GPS link format	Google Maps URL in SMS
System auto-reset cooldown	60 seconds
False positives	Zero (debounce verified)

12. Applications

- Agricultural irrigation systems
- Rural farm motor protection
- Streetlight cable protection
- Railway signaling systems
- Power distribution networks
- Industrial motor protection

13. Future Scope

- IoT-based remote monitoring dashboard for multi-farm management

- AI/ML-based theft prediction and anomaly detection
- Cloud integration for centralized analytics
- Dedicated mobile application for farmers
- Solar charging integration for fully off-grid deployment

14. Conclusive Discussion

FarmSecureX successfully addresses the critical issue of motor and copper theft in agriculture through its novel coded pulse continuity technique. The system has been live-tested and validated with actual hardware — demonstrating wire tamper detection in under 10 seconds, motor theft detection in under 1 second, successful SMS + voice call alerts, and high-quality UXGA photo evidence delivered via email. Designed to be low-cost, power-efficient, and scalable, the system is highly suitable for rural agricultural environments compared to CCTV or manual guards. A Utility Patent has been filed (Application No. 202521123445, filed 7th December 2025), confirming the novelty of the coded pulse approach. The system serves as a strong foundation for future enhancements including AI-based surveillance, cloud analytics, and mobile application support.

15. Work Plan and Timeline

Month	Tasks / Activities	Outcome
August	Literature survey, requirement analysis, component study	Problem understanding
September	Hardware selection, procurement, initial circuit design	Bill of materials + design
October	Hardware implementation, coding, communication testing	Working prototype
November	Integration of theft detection logic, SMS/call alert system	Functional system
December	Field testing, patent filing (filed 7th Dec 2025), optimization	Patent filed + validated
January	Documentation, GitHub repo, result screenshots, final demo	Complete project + demo

16. References

Journal Articles:

16. "Kottur farmers lament frequent theft of copper wire used for agricultural motor-pump sets in coconut groves," The Hindu, Coimbatore, India, Feb. 2023.
17. P. Thorat and R. N. Rathod, "Antitheft Alarm System for Electric Motor Pumps," IARJSET, NCETETE-2017, vol. 4, Special Issue 2, pp. 129-131, doi: 10.17148/IARJSET/NCETETE.2017.39
18. A. Sidebottom, M. Ashby, and S. D. Johnson, "Copper Cable Theft: Revisiting the Price-Theft Hypothesis," Journal of Research in Crime and Delinquency, vol. 51, no. 5, pp. 684-700, Aug. 2014, doi: 10.1177/0022427814521216.

GitHub Repository:

19. FarmSecureX Smart Surveillance System — <https://github.com/atharav-todkar/FarmSecureX-Smart-Surveillance-System>

Approval Signatures

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Guide

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